Customer Churn Analysis & Segmentation – Final Project Report

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EXECUTIVE SUMMARY

This report presents a comprehensive analysis of customer churn behaviour and segmentation for a telecommunications service provider. The study integrates two complementary machine learning approaches an Artificial Neural Network (ANN) model for churn prediction and K-Means clustering for customer segmentation to deliver both predictive and behavioural insights that can directly inform retention strategy.

The core business challenge addressed is customer attrition: identifying which customers are at risk of leaving and understanding the underlying reasons behind their decisions. A reactive approach to churn is costly and inefficient; this analysis aims to equip decision-makers with the tools to act proactively and with precision.

INTRODUCTION

In the telecommunications industry today, customers have more choices than ever. With many providers offering similar plans and services, switching has become quick and easy. Because of this, even small issues like high prices, poor network quality, or unsatisfactory service can lead customers to leave. As a result, keeping existing customers has become just as important, if not more important, than attracting new ones. Customer churn continues to be a key challenge for telecom companies, as it directly affects revenue and long-term business stability.

This project focuses on understanding customer churn specifically within a telecommunications context. Instead of relying on assumptions, it uses real customer data to explore what actually influences customers' decisions. Factors such as contract type, monthly charges, internet service type, and how long a customer has stayed with the company are especially relevant in telecom services and can strongly impact whether a customer stays or leaves.

To explore this, two main approaches are used. First, a predictive modelling approach using an Artificial Neural Network (ANN) is applied to identify customers who are more likely to churn. Second, customers are grouped into segments based on their characteristics and behaviour, which helps highlight patterns across different types of customers. Together, these approaches help answer two important questions: who is likely to leave, and what factors are driving that decision.

The goal of this report is to turn these insights into practical actions. By understanding customers more clearly and identifying risks earlier, telecom companies can take more targeted steps to improve customer experience, reduce churn, and build stronger relationships over time.

1. FACTORS CONTRIBUTING TO CHURN AND RETENTION

The type of contract a customer holds is one of the most significant factors influencing churn. Customers on month-to-month contracts exhibit substantially higher churn rates compared to those on one-year or two-year agreements. This pattern demonstrates a direct and meaningful correlation between contract flexibility and churn risk: the shorter and more flexible the commitment, the easier it is for a customer to leave.

Customers on short-term contracts face fewer switching barriers. They are more exposed to and responsive to competitor offers and are more likely to leave when experiencing even minor service dissatisfaction. They have, in effect, opted for maximum freedom and that freedom makes them structurally harder to retain.

Long-term contracts, by contrast, are associated with significantly higher customer stability. These agreements cultivate loyalty, often include incentive structures that reward continued engagement, and create meaningful switching costs. Contract architecture is therefore a powerful lever in customer retention. The business must actively work to migrate customers from flexible to committed arrangements, using financial or service-based incentives to ease the transition.

1.1 INTERNET SERVICE TYPE AS THE STRONGEST SEGMENTATION FACTOR

Among all variables tested, internet service type emerged as the most significant factor in customer segmentation. This finding highlights the central role that service configuration plays in determining behavioural variation across customer groups. Different service types, notably fibre optic versus DSL, differ in performance expectations, pricing, reliability, and the quality of customer experience they deliver.

Customers subscribed to certain service types may have higher baseline expectations for speed, reliability, and value-for-money. When these expectations are not consistently met, whether through outages, speed drops, or pricing misalignment, dissatisfaction escalates rapidly and churn risk grows. The type of internet service, therefore, acts as both a behavioural differentiator and a satisfaction driver, directly shaping how customers use the service, how they perceive its value, and the strength of their loyalty.

From a segmentation perspective, internet service type proved more powerful than demographics in explaining how customers naturally group themselves. This reinforces the need for service-quality-led retention strategies rather than demographic profiling.

1.2 MONTHLY CHARGES AND PRICING PRESSURE

Monthly charges represent another highly significant churn driver, particularly through the mechanism of pricing pressure. Customers with higher monthly fees are more likely to leave a pattern that is especially pronounced in the high-value volatile cluster identified through the clustering analysis (Cluster 3). This creates a paradox: the customers who generate the most revenue are also the most at risk.

Higher-paying customers tend to hold elevated expectations. They expect superior service quality, faster response times, and overall value commensurate with what they are paying. When these expectations are not consistently fulfilled, dissatisfaction grows and the probability of churn increases. It is important to note, however, that pricing alone cannot fully explain churn behaviour there remains overlap between churned and retained customers at similar price points. This suggests that price sensitivity operates in conjunction with other factors such as contract type, service quality, and perceived value, rather than acting in isolation.

The implication for strategy is clear: pricing optimisation must be accompanied by service quality improvements. Reducing the gap between what customers pay and what they perceive they receive is essential to retaining high-value segments.

1.3 TENURE AND ITS RELATIONSHIP WITH CHURN

Customer tenure defined as the length of time a customer has been with the service provider shows a strong negative correlation with churn: the longer a customer remains, the less likely they are to leave. This is an intuitive finding consistent with loyalty literature, as customers who have invested time with a provider are more likely to have adapted their usage patterns, built service familiarity, and experienced the accumulated benefits of staying.

However, while tenure is a useful indicator of customer stability, it does not have a strong impact on clustering outcomes. This means that tenure alone is insufficient to segment customers or fully explain their behaviour. Churn risk among newer customers is significantly higher, and the findings suggest that the critical window for retention intervention is the early stages of the customer lifecycle particularly the first two months, before habits and loyalty have had time to form.

Customers classified as 'Noncustomers', those who have been with the provider for two months or less, display the least consistent behavioural patterns and represent a particularly vulnerable group. This early-stage churn can have a disproportionately negative impact on overall retention rates if not proactively managed through structured onboarding and engagement programmes.

1.4 DEMOGRAPHIC FACTORS AND THEIR LIMITED INFLUENCE

A significant and practically important finding of this analysis is that demographic variables, including gender and senior citizen status, had minimal bearing on customer segmentation or churn prediction. This challenges the traditional assumption that demographics are reliable proxies for customer behaviour.

In this dataset, demographic characteristics alone are not reliable predictors of churn. Instead, behavioural and service-based criteria, specifically contract type, service configuration, pricing, and usage patterns, are far more powerful explanatory variables. This finding has direct implications for how the business designs its retention strategies: demographic profiling should be deprioritised in favour of behavioural segmentation. Interventions tailored to how customers use and relate to the service will be more targeted, more effective, and more respectful of individual customer variation.

2. SUMMARY OF KEY FINDINGS

2.1 OVERVIEW

This analysis combined Artificial Neural Network (ANN) modelling with K-Means clustering to examine customer churn and identify meaningful customer segments. The purpose was to not only predict which customers are likely to churn, but also to understand the behavioural patterns behind different customer groups. By integrating these two approaches, the analysis provides both predictive insights who will churn and segmentation insights why they behave differently. This allows for more informed and targeted decision-making when developing customer retention strategies.

2.2 KEY FINDINGS FROM CHURN PREDICTION (ANN MODEL)

The ANN model was trained on a fully pre-processed dataset, where features were scaled and prepared to ensure effective learning. The model used multiple hidden layers to capture complex, non-linear relationships between customer attributes and churn behaviour.

BASELINE MODEL PERFORMANCE

The baseline ANN model achieved an overall accuracy of approximately 76%, with a churn recall of 41%. While this indicates reasonable general performance, the low recall figure reveals a critical weakness: the model initially failed to identify the majority of customers who actually churned. In a business context, these are missed intervention opportunities customers who left without being flagged for retention action. This underperformance on the minority class is a common challenge in imbalanced classification problems, and it underscores why overall accuracy is an insufficient measure of model quality in churn prediction tasks.

TUNED MODEL: THRESHOLD ADJUSTMENT AND ARCHITECTURAL IMPROVEMENTS

To address the baseline model's shortcomings, the model was tuned by adjusting its architecture and modifying the classification threshold from 0.5 to 0.25. Lowering the threshold means the model classifies a customer as a churn risk at a lower predicted probability making it more sensitive to at-risk cases. The result was a significant and meaningful improvement: churn recall rose to 76%, meaning the model now correctly identifies approximately three out of every four customers who will actually leave.

The trade-off is a slight reduction in overall accuracy, which fell from 76% to 71%. This is expected and acceptable: by widening the net to catch more churners, the model inevitably flags some non-churners incorrectly (false positives). In a business setting, the cost of a false positive engaging a loyal customer with a retention offer is far lower than the cost of a false negative, which means losing a customer entirely without warning.

FIGURE 1: BASELINE VS. TUNED ANN MODEL PERFORMANCE

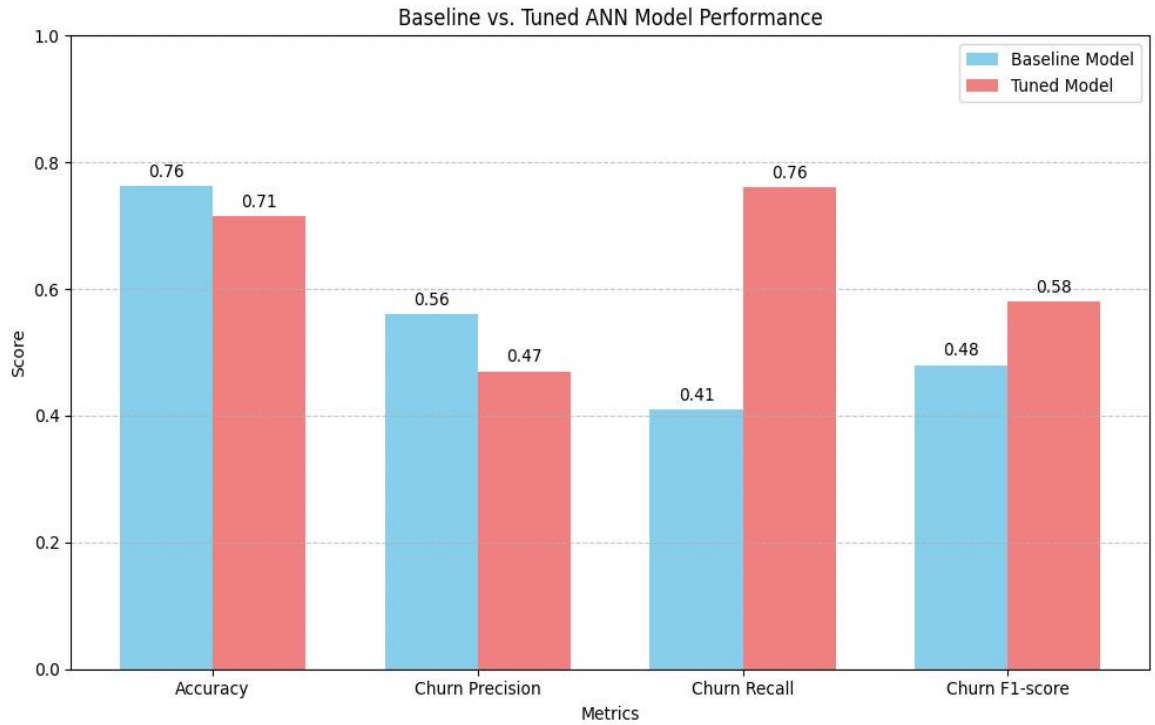

Figure 1: Comparison of Baseline and Tuned ANN Model Performance across Accuracy, Churn Precision, Churn Recall, and Churn F1-Score

The bar chart presents a side-by-side comparison of four key performance metrics for both the baseline model (light blue) and the tuned model (salmon/red) across Accuracy, Churn Precision, Churn Recall, and Churn F1-Score.

Accuracy: The baseline model scores 0.76, while the tuned model scores 0.71 a modest decrease of 5 percentage points. This reflects the deliberate trade-off of catching more churners at the cost of slightly more false positives overall.

Churn Precision: The baseline model scores 0.56 versus the tuned model's 0.47. The lower precision of the tuned model confirms that it flags more non-churners as at-risk consistent with the lower threshold setting.

Churn Recall: The most critical metric. The tuned model dramatically outperforms the baseline at 0.76 versus 0.41 an improvement of 35 percentage points. This is the headline outcome of the tuning process, representing a near-doubling of the model's ability to detect actual churners.

Churn F1-Score: The F1-score, which balances precision and recall, rises from 0.48 to 0.58 with tuning. This improvement confirms that the tuned model achieves a better overall balance and is more effective in real-world deployment, where both false positives and false negatives carry costs.

The visual contrast between the two models is stark on the Recall axis the tuned model's salmon bar towers above the baseline, while the differences on Accuracy and Precision are smaller, reinforcing that the tuning was well-calibrated for the business objective.

FIGURE 2: CONFUSION MATRIX FOR THE TUNED ANN MODEL (THRESHOLD: 0.25)

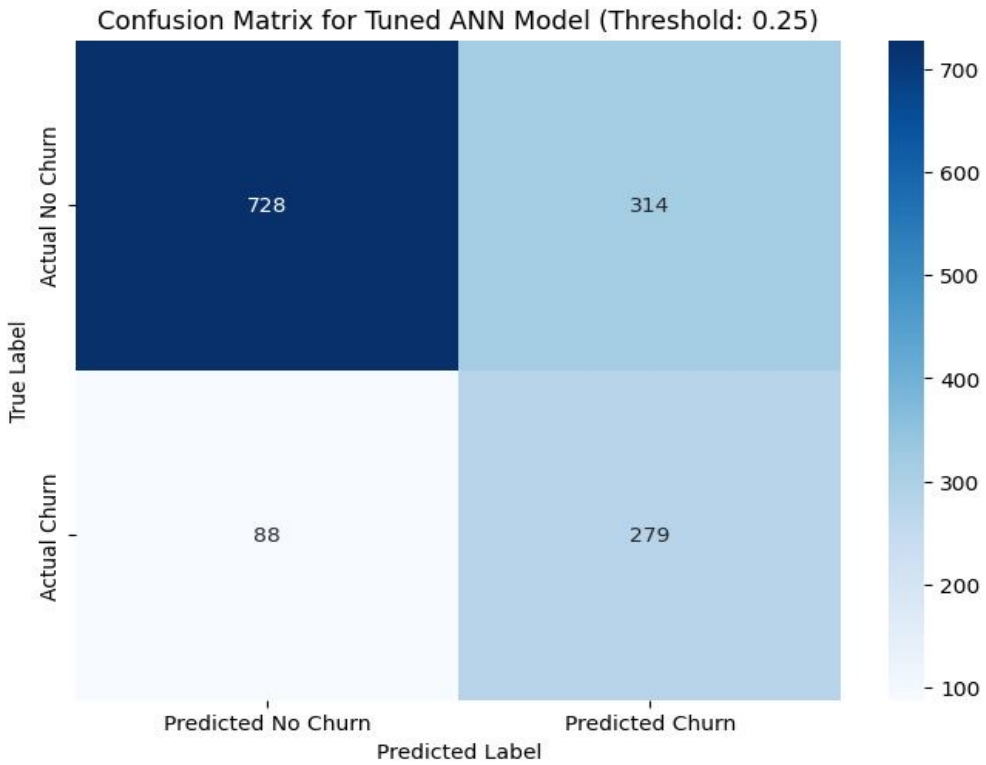

Figure 2: Confusion Matrix for the Tuned ANN Model at Classification Threshold of 0.25

The confusion matrix provides a granular view of model predictions, broken into four quadrants:

- True Negatives (728) top-left, dark navy: The model correctly identified 728 customers as non-churners who did not churn. This represents sound baseline performance on the majority class.
- False Positives (314) top-right, light blue: The model incorrectly flagged 314 non-churning customers as likely to churn. These are the cost of a lower threshold customers who will receive retention offers they technically do not need, but this is operationally manageable.
- False Negatives (88) bottom-left, near-white: Only 88 actual churners were missed by the model. Compared to the baseline model's much higher miss rate, this represents a substantial improvement and means far fewer customers slip through undetected.
- True Positives (279) bottom-right, light blue: The model correctly identified 279 customers who actually churned. This is the primary measure of practical utility these customers can be targeted for retention intervention before they leave.

The matrix visually reinforces the tuned model's strength: the large dark block in the top-left (true negatives) and the meaningful volume of correctly predicted churners (279 true positives) against only 88 missed cases. The colour gradient from deep navy to light blue corresponds to frequency of prediction, making cluster patterns immediately readable.

2.3 KEY FINDINGS FROM CUSTOMER SEGMENTATION

Customer segmentation was performed using K-Means clustering to group customers based on similar characteristics and behaviours. The Elbow Method was used to determine the optimal number of clusters by examining the rate at which clustering performance improves as the number of clusters increases.

The Elbow Method analysis showed that the improvement in clustering performance slows significantly after four clusters, indicating that four groups provide the optimal balance between simplicity and meaningful segmentation. Dividing customers into fewer groups loses important behavioural distinctions, while more clusters produces diminishing returns in explanatory power.

The clustering results revealed that contract type and service-related features are the most important factors distinguishing customer groups. Demographic variables such as gender had minimal impact on how customers were segmented. This confirms that customer behaviour is driven more by how services are structured and consumed, rather than by who the customer demographically is.

CUSTOMER SEGMENT INSIGHTS

The four clusters represent distinct customer groups with clearly differentiated behaviours and churn risks:

- Cluster 0 Stable Loyalists: These customers exhibit high average tenure and moderate monthly charges, with a churn rate of 25.36%. They represent the most stable segment long-term customers with consistent service usage patterns who are generally satisfied, loyal, and less reactive to pricing or competitive pressures.
- Cluster 1 Standard Users: Similar in profile to Cluster 0, with high tenure, moderate charges, and a churn rate of 26.17%. These customers show mixed service usage but are fundamentally stable. They represent a reliable base that can be cultivated through loyalty programmes and targeted upselling.
- Cluster 2 Emerging Customers: Characterised by moderate tenure and moderate monthly charges, with a churn rate of 25.79%. These are customers in the early or transitional stages of their relationship with the provider, whose behavioural patterns have not yet stabilised. Early-stage engagement is critical for this group.
- Cluster 3 High-Value Volatile: Moderate tenure but the highest average monthly charges (~0.47), with the highest churn rate of 28.73%. This is the most strategically critical segment customers who generate the most revenue but are most at risk of leaving. Their instability is linked to pricing sensitivity and the flexibility of month-to-month contracts.

3. RECOMMENDATIONS FOR TARGETED RETENTION STRATEGIES

3.1 OVERVIEW

To increase customer retention and maximise overall customer lifetime value, a set of focused retention tactics has been designed based on insights obtained from customer segmentation and churn factor research. These recommendations are customised for each consumer segment identified through clustering, as opposed to employing a standard approach for the entire customer base. Each segment receives interventions that align with its behavioural characteristics, degree of churn risk, and business value. By connecting data-driven insights to practical tactics, the company can more effectively reduce attrition and strengthen long-term client relationships.

3.2 DEEP CLUSTER PROFILING

A deeper analysis of cluster behaviour revealed clear and actionable differences between customer groups. The most influential variables shaping these clusters were internet service type, contract type (especially month-to-month versus long-term contracts), tenure, and monthly charges. Notably, demographic variables such as gender and senior citizen status had very little impact on clustering, confirming that customer behaviour is a more powerful lens than demographics for understanding group patterns.

CLUSTER PROFILE SUMMARY TABLE

Cluster	Customer Type	Avg Tenure	Avg Monthly Charges	Churn Rate	Key Behaviour
Cluster 0	Stable Loyalists	High (~0.45)	Moderate (~0.47)	25.36%	Long-term, stable customers
Cluster 1	Standard Users	High (~0.45)	Moderate (~0.45)	26.17%	Mixed service usage, stable
Cluster 2	Emerging Customers	Moderate (~0.44)	Moderate (~0.45)	25.79%	Early-stage customers
Cluster 3	High-Value Volatile	Moderate (~0.44)	Highest (~0.47)	28.73%	High-paying but unstable

Table 1: Cluster Profile Summary Customer Segments by Behaviour and Churn Risk

3.3 PRIORITY 1: CLUSTER 3 HIGH-VALUE VOLATILE CUSTOMERS

From a business standpoint, Cluster 3 is the most strategically important segment. These customers generate the highest monthly revenue but simultaneously display the highest churn rate of 28.73%. They are typically enrolled in month-to-month contracts and are paying the highest

charges a combination that makes them acutely sensitive to competitive offers and perceived service dissatisfaction. Despite their financial significance, their lack of contractual commitment means they can and do leave with minimal friction.

RECOMMENDED STRATEGY: CONTRACT CONVERSION WITH INCENTIVES

A targeted contract conversion strategy is the primary recommended intervention for this segment. The goal is to encourage customers to migrate from flexible, short-term contracts to longer-term agreements specifically one-year or two-year plans. This structural shift removes the ease of departure and creates meaningful retention anchors.

To make this transition attractive, monetary incentives should be offered: a discount of approximately 15% on the annual plan price provides a tangible financial benefit that justifies the switch. Beyond price incentives, the conversion should be supported by value-added features bundled services, preferential service tiers, and loyalty reward programmes that increase the perceived value of committing to the provider long term. These additional benefits raise switching costs by making the service ecosystem richer and harder to replicate elsewhere.

The anticipated outcomes of this strategy include increased customer commitment to longer-term agreements, a measurable decrease in churn rate within this segment, and protection of the business's highest-revenue customer cohort. By stabilising Cluster 3, the business safeguards its most significant revenue stream.

3.4 PRIORITY 2: CLUSTER 2 EMERGING CUSTOMERS

Cluster 2 consists of customers in the early or transitional phases of their relationship with the service provider. These customers have moderate tenure and display behavioural patterns that have not yet stabilised they are still forming their habits, evaluating the service's value proposition, and deciding whether to commit long term. Their moderate churn risk reflects this uncertainty: without positive reinforcement and structured engagement, many will not progress to becoming loyal, long-term customers.

RECOMMENDED STRATEGY: CUSTOMER NURTURING AND ONBOARDING ENHANCEMENT

A customer nurturing strategy is recommended for this segment, with particular focus on improving the quality and warmth of the early customer experience. Proactive engagement should begin immediately upon customer acquisition: personalised welcome communications, structured onboarding journeys, instructional materials (including service tutorial videos and usage guides), and responsive support channels should all be deployed within the first weeks of the customer relationship.

These initiatives serve two purposes. First, they help customers understand and fully utilise the features and benefits of the service, increasing perceived value and satisfaction. Second, they signal to the customer that the provider is invested in their success building relational equity that makes future departure feel less attractive.

The goal is to make the service 'sticky' early: to create habits, dependencies, and positive associations that carry the customer through the transitional period and into a stable, long-term relationship. Anticipated outcomes include reduced early-stage churn, higher long-term retention rates, and the eventual conversion of emerging customers into the stable loyalist segments (Clusters 0 and 1).

3.5 PRIORITY 3: CLUSTERS 0 AND 1 STABLE CUSTOMERS

Clusters 0 and 1 represent the most stable and loyal customer segments in the portfolio. These customers have longer tenure, moderate consumption habits, and low churn rates indicating a high baseline of satisfaction and engagement. While their retention risk is lower, this does not mean they should be taken for granted. In fact, these segments offer significant opportunities for value expansion through upselling, cross-selling, and advocacy activation.

RECOMMENDED STRATEGY: LOYALTY PROGRAMME AND UPSELLING

A strategy centred on expansion and loyalty reinforcement is recommended for these clusters. VIP loyalty programmes should be designed to formally reward the commitment these customers have already demonstrated. Programme benefits could include early access to new features, exclusive service upgrades, priority customer support, and referral bonuses that incentivise these loyal customers to become active brand advocates.

Additionally, targeted upselling should be explored specifically offering premium services or add-ons that are curated to match individual customer usage profiles and needs. This approach simultaneously increases revenue per customer and deepens the service relationship, making it more embedded in the customer's daily routine and therefore harder to abandon.

Anticipated outcomes include enhanced engagement across these segments, improved customer lifetime value, and the cultivation of brand champions who drive word-of-mouth acquisition.

3.6 CROSS-SEGMENT STRATEGIES

In addition to cluster-specific interventions, several cross-segment strategies should be implemented to enhance overall retention performance:

PRICING OPTIMISATION

Pricing structures should be reviewed and realigned to better match customer perceptions of value. This includes introducing flexible pricing tiers that allow customers to scale their commitment and investment in line with their needs. The key objective is to close the gap between what customers pay and what they believe they are receiving because it is this gap, not price level alone, that drives dissatisfaction and departure.

ANN-BASED CHURN MONITORING

The ANN model should be deployed as a continuous operational tool, not merely an analytical exercise. By running the model at regular intervals on live customer data, the business can identify customers crossing into high-risk churn territory before they leave. This enables proactive, well-timed retention interventions tailored communications, personalised offers, or direct outreach delivered precisely when they are most likely to be effective. Automating this process within a CRM or data pipeline significantly increases scalability and efficiency.

SERVICE PERSONALISATION

Across all segments, service personalisation represents a powerful but underutilised lever. By analysing individual usage patterns and preferences, the business can tailor service plans, feature recommendations, and communications to individual needs. This personalised approach strengthens the emotional and practical bond between customer and provider, improving satisfaction and reducing the appeal of switching to a competitor offering a generic alternative.

4. LIMITATIONS AND PROPOSED SOLUTIONS

4.1 OVERVIEW

Throughout the project, several limitations were identified that affect the robustness, depth, and practical applicability of the findings. These limitations relate to model performance, data quality, feature coverage, interpretability, and evaluation methodology. Acknowledging these challenges transparently is essential to ensuring the analysis is used appropriately and to charting a clear path for future improvement. For each limitation identified, a concrete proposed solution is outlined below.

4.2 MODERATE CLUSTER SEPARATION

A silhouette score of 0.192 was achieved in the clustering analysis, indicating a modest degree of cluster separation. This score confirms the existence of meaningful groupings within the data, but also reveals that there is overlap between clusters meaning that the boundaries between customer segments are not sharply defined. In real-world behavioural datasets, where customer attributes exist along continuous spectra rather than in distinct categories, some degree of overlap is normal and expected. However, it does limit the precision and clarity of segment boundaries, which in turn may reduce the specificity of targeted interventions.

PROPOSED SOLUTION

Alternative clustering algorithms capable of capturing more complex, non-linear relationships should be explored. DBSCAN (Density-Based Spatial Clustering of Applications with Noise) can identify clusters of arbitrary shape and naturally handles outliers, making it well-suited to behavioural data. Hierarchical clustering offers another perspective, allowing the examination of cluster merges at multiple levels of granularity. Additionally, enriching the feature set with more behavioural variables usage frequency, service interaction patterns, complaint history may produce more distinctly separated clusters and more actionable segment profiles.

4.3 LIMITED FEATURE SET

The current dataset, while informative, does not capture some crucial dimensions of customer behaviour and experience. Specifically, customer satisfaction scores, complaint histories, direct feedback, and qualitative sentiment data are absent. These are precisely the variables that often explain the 'why' behind churn decisions the subjective experience that motivates a customer to leave and their absence limits the model's explanatory depth.

PROPOSED SOLUTION

Future iterations of the analysis should prioritise feature enrichment. Customer feedback data including survey responses, Net Promoter Scores (NPS), and satisfaction ratings can provide rich qualitative signals. Service interaction logs, including usage volume, call centre contact frequency, and support ticket history, can add behavioural texture. By expanding the feature set to include these dimensions, the model can develop a more complete and nuanced understanding of the customer experience, improving both prediction accuracy and segmentation meaningfulness.

4.4 ANN MODEL INTERPRETABILITY

Artificial Neural Networks are powerful tools for capturing complex, non-linear patterns in data, but they are inherently difficult to interpret. The internal mechanics of an ANN how specific input features combine across multiple hidden layers to produce a prediction are not transparently readable by humans. This 'black box' quality is a meaningful limitation in a business context, where stakeholders need to understand and trust the model's outputs in order to act on them confidently. Without interpretability, there is a risk that the model is used without adequate critical oversight.

PROPOSED SOLUTION

Explainability frameworks should be integrated into the modelling pipeline. SHAP (SHapley Additive exPlanations) values provide a rigorous, mathematically grounded method for attributing each feature's contribution to individual predictions making it possible to say not just 'this customer is at high risk' but 'this customer is at high risk primarily because of their contract type and monthly charges.' LIME (Local Interpretable Model-Agnostic Explanations) offers a complementary approach for local explanations. Benchmarking the ANN against more inherently interpretable models logistic regression, decision trees can also provide a useful reference baseline, helping stakeholders understand where the ANN adds value through complexity and where simpler models perform comparably.

4.5 STATIC DATASET LIMITATION

The analysis is based on a static historical snapshot of customer data a dataset that represents customer behaviour at a specific point in time. Consumer behaviour, however, is dynamic: it shifts in response to market conditions, competitor pricing changes, macroeconomic pressures, service updates, and seasonal patterns. A model trained on static data will inevitably become less accurate over time as the distribution of customer behaviour drifts away from what was observed in training.

PROPOSED SOLUTION

A dynamic data pipeline should be implemented to enable regular model retraining on fresh data. Ideally, this would involve automated data collection, preprocessing, and model retraining at regular intervals monthly or quarterly with performance metrics monitored continuously to detect model drift. In more advanced implementations, a real-time or near-real-time scoring pipeline would allow the churn model to continuously update its risk assessments as new customer behaviour data arrives, ensuring predictions remain current and actionable.

4.6 CLASS IMBALANCE

The dataset exhibits class imbalance in the target variable: non-churning customers substantially outnumber churning customers. In such situations, machine learning models tend to be biased toward predicting the majority class in this case, non-churn because doing so maximises overall accuracy metrics without genuinely learning to identify churn patterns. The consequence is that the model may achieve high headline accuracy while performing poorly on the minority class (actual churners), which is precisely the class of most business interest.

PROPOSED SOLUTION

The Synthetic Minority Over-sampling Technique (SMOTE) can be applied to generate synthetic churn examples, effectively balancing the class distribution in the training data without simply duplicating existing records. Alternatively, or in combination with SMOTE, class weighting can be

applied during model training to impose a higher penalty on misclassifying churn cases making the model more sensitive to the minority class. Both approaches directly address the imbalance and improve the model's ability to detect at-risk customers accurately.

4.7 LIMITED VALIDATION DEPTH

The current evaluation of the ANN model relies primarily on accuracy and recall metrics. While these are informative, they do not provide a complete picture of model performance across all classification thresholds and edge cases. Without a broader evaluation framework, it is difficult to fully characterise the model's strengths and weaknesses or to compare it rigorously against alternative approaches.

PROPOSED SOLUTION

A comprehensive evaluation framework should be adopted, incorporating the full suite of relevant metrics: precision, recall, F1-score, ROC-AUC (Receiver Operating Characteristic Area Under Curve), and PR-AUC (Precision-Recall Area Under Curve). Cross-validation methodologies specifically k-fold cross-validation should be applied to reduce variance in performance estimates and provide more reliable generalisability assessments. The ROC-AUC statistic is particularly valuable as it measures the model's discriminative ability across all possible classification thresholds, providing a threshold-independent view of performance.

5. SUMMARY AND CONCLUSION

5.1 SUMMARY OF FINDINGS

This report has presented a comprehensive, multi-method analysis of customer churn and segmentation for a telecommunications provider. By combining the predictive power of an Artificial Neural Network model with the descriptive insights of K-Means clustering, the analysis delivers a richly layered understanding of customer behaviour that neither approach could achieve alone.

The ANN churn prediction model demonstrates strong operational potential. Following threshold tuning, the model achieves a churn recall of 76% correctly identifying approximately three out of every four customers who would leave while maintaining a practical level of overall accuracy at 71%. The confusion matrix confirms that the model now captures 279 true positive churn cases while missing only 88, a dramatic improvement over the baseline's performance. The deliberate trade-off between precision and recall is well-calibrated to the business objective: in churn prevention, it is far more costly to miss a churner than to over-extend a retention offer.

The segmentation analysis identified four distinct and actionable customer clusters. Clusters 0 and 1 represent stable, loyal customers with moderate churn risk who offer opportunities for loyalty reinforcement and value expansion. Cluster 2 encompasses emerging customers in a critical transitional period, where the quality of early experience determines long-term retention outcomes. Cluster 3 High-Value Volatile customers represents the most urgent business priority: high-revenue customers whose combination of elevated monthly charges, flexible contracts, and price sensitivity creates the highest churn risk in the portfolio.

The factors analysis confirmed that contract type, internet service type, and monthly charges are the dominant drivers of churn, while demographic variables are largely irrelevant to churn prediction. Customer tenure, while not a strong segmentation variable, remains a meaningful protective factor longer-tenured customers are substantially more stable reinforcing the importance of early-stage retention investment.

5.2 OVERALL KEY INSIGHT

The overarching insight of this analysis is that successful customer retention must be grounded in behavioural segmentation rather than demographic generalisation. Customer behaviour how services are used, which contracts are chosen, how sensitive individuals are to pricing provides a more precise and actionable foundation for decision-making than traditional demographic profiles.

Furthermore, the combination of ANN modelling and clustering is greater than the sum of its parts. The ANN identifies who is at risk; the clustering explains why different groups of customers behave as they do. Together, they provide the strategic clarity needed to design targeted, proportionate, and evidence-based retention interventions. Organisations that act on behavioural insights rather than demographic assumptions, and that deploy predictive models continuously rather than periodically, will be significantly better positioned to reduce attrition and improve customer lifetime value.

5.3 CONCLUSION

In conclusion, this analysis demonstrates that customer churn is a predictable and addressable business challenge, driven by a combination of behavioural and service-related factors rather

than by demographics alone. The ANN model proved highly effective in predicting churn after threshold tuning, while K-Means clustering revealed four distinct customer segments with meaningfully different risk profiles and retention needs.

The recommendations provided targeted contract conversion for high-value volatile customers, structured onboarding for emerging customers, loyalty programmes for stable segments, and cross-cutting pricing optimisation and ANN-based monitoring offer a coherent and evidence-based roadmap for reducing churn and strengthening long-term customer relationships.

Acknowledged limitations including moderate cluster separation, a static dataset, limited feature richness, and ANN interpretability challenges do not undermine the value of the findings but identify clear opportunities for future refinement. Implementing the proposed solutions richer data collection, advanced clustering algorithms, SHAP-based explainability, and dynamic model retraining will significantly enhance the robustness and business impact of subsequent analyses.

Ultimately, the value of this work lies not in the models themselves, but in the actionable intelligence they generate. By translating data into targeted strategy, this analysis equips the business to move from reactive churn management to proactive, precision retention protecting revenue, deepening customer relationships, and building a more resilient customer base over time.

End of Report